THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** -4	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
2	**(b)** 2	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
3	**(a)** 8	Solve for x and y
4	**(b)** 12	Intercepts 6 and 4; area $= \frac{1}{2} \times 6 \times 4$
5	**(c)** 29	$(x+y)^2 - 2xy = 49 - 20$
6	**(a)** 3	Subtract the equations
7	**(b)** -6	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
8	**(a)** 14	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$3 + 1 = 4$ and $3 - 1 = 2$; the common point satisfies both equations, which is exactly the solution (R).
10	**(c)** A is true but R is false.	$\frac{1}{2} = \frac{2}{4} \neq \frac{4}{12}$, so the lines are parallel and there is no solution: A true; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{k}{8} = \frac{2}{k} \Rightarrow k = \pm 4$ (1); $k = 4$ gives coincident lines, so $k = -4$ (1).	1+1
12	Adding: $8(x+y) = 88$ so $x + y = 11$ (1); subtracting: $x - y = 3$, so $x = 7$, $y = 4$ (1).	1+1
13	x-intercept $(4, 0)$, y-intercept $(0, 6)$ (1); area $= \frac{1}{2} \times 4 \times 6 = 12$ (1).	1+1
14	$\frac{k}{12} = \frac{3}{k} \Rightarrow k = \pm 6$ (1); $k = 6$ gives coincident lines, so $k = -6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $2p + 3q = 2$, $3p + 5q = \frac{19}{6}$ (1); eliminating q: $(1)p = \frac{1}{2}$ so $p = \frac{1}{2}$ (1); $q = \frac{1}{3}$ (1); $x = 2$, $y = 3$ (1).	1+1+1+1
16	Need $\frac{a+b}{3} = \frac{a-b}{1} = \frac{14}{7} = 2$ (1); $a + b = 6$, $a - b = 2$ (1); $a = 4$, $b = 2$ (1).	1+1+1
17	Intersection $(4, 2)$ (1); x-intercepts 6 and 2 (1); area $= \frac{1}{2} \times 4 \times 2 = 4$ (1).	1+1+1
	OR: Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $3p + 4q = \frac{11}{4}$, $5p + 2q = \frac{9}{4}$ (1); eliminating q: $(-14)p = -\frac{7}{2}$ so $p = \frac{1}{4}$ (1); $q = \frac{1}{2}$ (1); $x = 4$, $y = 2$ (1).	1+1+1+1
18	Need $\frac{a+b}{5} = \frac{a-b}{3} = \frac{12}{4} = 3$ (1); $a + b = 15$, $a - b = 9$ (1); $a = 12$, $b = 3$ (1).	1+1+1
	OR: Intersection $(3, 2)$ (1); x-intercepts 4 and 5 (1); area $= \frac{1}{2} \times 1 \times 2 = 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3n + 2p = 70$ and $2n + 3p = 65$ (1). (ii) Multiply by 3 and 2: $9n + 6p = 210$, $4n + 6p = 130$; subtract: $5n = 80$, so $n = ₹\,16$ (1). (iii) $3(16) + 2p = 70 \Rightarrow p = ₹\,11$ (1). (iv) 6 notebooks cost ₹ 96; remaining ₹ 104; $104 \div 11 = 9$ remainder 5, so 9 pens (1) and ₹ 5 left (1).	1+1+1+2
20	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $21p + 33q = 10$, $12p + 55q = 9$ (2); $p = \frac{1}{3}$, $q = \frac{1}{11}$ (1); $x - y = 3$, $x + y = 11$ (1); boat 7 km/h, stream 4 km/h (1).	2+1+1+1
	OR: Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $480p + 420q = \frac{51}{5}$, $600p + 300q = \frac{21}{2}$ (2); $p = \frac{1}{80}$, $q = \frac{1}{100}$ (2); train 80 km/h, bus 100 km/h (1).	2+2+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.

- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 41	$(x+y)^2 - 2xy = 81 - 40$
2	**(a)** $\frac{7}{3}$	Subtract the equations
3	**(a)** -8	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
4	**(b)** 16	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
5	**(b)** 9	Solve for x and y
6	**(a)** 20	Intercepts 4 and 10; area $= \frac{1}{2} \times 4 \times 10$
7	**(a)** 58	$(x+y)^2 - 2xy = 100 - 42$
8	**(b)** 2	Subtract the equations
9	**(d)** A is false but R is true.	$\frac{3}{2} \neq \frac{2}{-3}$, so the lines intersect (not parallel): A false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$3 + 2 = 5$ and $3 - 2 = 1$, so A is true; R is also true but it does not verify A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adding: $11(x+y) = 115$ so $x + y = \frac{115}{11}$ (1); subtracting: $x - y = \frac{7}{3}$, so $x = \frac{211}{33}$, $y = \frac{134}{33}$ (1).	1+1
12	x-intercept $(5, 0)$, y-intercept $(0, 4)$ (1); area $= \frac{1}{2} \times 5 \times 4 = 10$ (1).	1+1
13	$\frac{k}{16} = \frac{4}{k} \Rightarrow k = \pm 8$ (1); $k = 8$ gives coincident lines, so $k = -8$ (1).	1+1
14	Adding: $8(x+y) = 68$ so $x + y = \frac{17}{2}$ (1); subtracting: $x - y = 2$, so $x = \frac{21}{4}$, $y = \frac{13}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $2p + 5q = \frac{33}{20}$, $3p + 2q = \frac{11}{10}$ (1); eliminating q: $(-11)p = -\frac{11}{5}$ so $p = \frac{1}{5}$ (1); $q = \frac{1}{4}$ (1); $x = 5$, $y = 4$ (1).	1+1+1+1
16	Need $\frac{a+b}{7} = \frac{a-b}{1} = \frac{10}{5} = 2$ (1); $a + b = 14$, $a - b = 2$ (1); $a = 8$, $b = 6$ (1).	1+1+1
17	Intersection $(2, 3)$ (1); x-intercepts 8 and $\frac{7}{2}$ (1); area $= \frac{1}{2} \times \frac{9}{2} \times 3 = \frac{27}{4}$ (1).	1+1+1
	OR: Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $4p + 3q = \frac{11}{6}$, $2p + 5q = \frac{3}{2}$ (1); eliminating q: $(14)p = \frac{14}{3}$ so $p = \frac{1}{3}$ (1); $q = \frac{1}{6}$ (1); $x = 3$, $y = 6$ (1).	1+1+1+1
18	Need $\frac{a+b}{4} = \frac{a-b}{2} = \frac{9}{3} = 3$ (1); $a + b = 12$, $a - b = 6$ (1); $a = 9$, $b = 3$ (1).	1+1+1
	OR: Intersection $(2, 3)$ (1); x-intercepts 3 and 5 (1); area $= \frac{1}{2} \times 2 \times 3 = 3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{2}{4} = \frac{3}{6} = \frac{12}{24}$: the roads coincide (1). (ii) $\frac{1}{2} = \frac{1}{2} \neq \frac{5}{7}$: parallel, never meet (1). (iii) Adding: $2x = 8$, $x = 4$, $y = 3$; junction $(4, 3)$ (1). (iv) Coincident when $\frac{k}{4} = \frac{3}{6}$, so $k = 2$ (1); for $k \neq 2$ the ratios $\frac{k}{4} \neq \frac{3}{6}$ and they meet at exactly one point (1).	1+1+1+2
20	Let boat x km/h, stream y km/h; put $p = \frac{1}{x-y}$, $q = \frac{1}{x+y}$: $45p + 60q = 13$, $30p + 90q = 12$ (2); $p = \frac{1}{5}$, $q = \frac{1}{15}$ (1); $x - y = 5$, $x + y = 15$ (1); boat 10 km/h, stream 5 km/h (1).	2+1+1+1
	OR: Intersection $(4, 2)$ (2); y-intercepts $(0, 10)$ and $(0, -2)$ (1); base $= 12$, height $= 4$ (1); area $= \frac{1}{2} \times$ base $\times$ height $= 24$ sq. units (1).	2+1+1+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.

- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
- Writing the reversed number as $10x + y$ again instead of $10y + x$ in digit problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 7	Solve for x and y
2	**(c)** 10	Intercepts 5 and 4; area $= \frac{1}{2} \times 5 \times 4$
3	**(a)** 74	$(x+y)^2 - 2xy = 144 - 70$
4	**(a)** 2	Subtract the equations
5	**(c)** -12	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
6	**(a)** 20	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
7	**(a)** 11	Solve for x and y
8	**(d)** 9	Intercepts 3 and 6; area $= \frac{1}{2} \times 3 \times 6$
9	**(a)** Both A and R are true and R is the correct explanation of A.	For $k = -6$: $\frac{-6}{12} = \frac{3}{-6} = -\frac{1}{2}$ but $\frac{3}{6} = \frac{1}{2}$; this is exactly the condition in R.
10	**(c)** A is true but R is false.	Need $\frac{2}{4} = \frac{1}{2} = \frac{5}{k} \Rightarrow k = 10$: A true. R is false: a pair has 0, 1 or infinitely many solutions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	x-intercept $(4, 0)$, y-intercept $(0, 6)$ (1); area $= \frac{1}{2} \times 4 \times 6 = 12$ (1).	1+1
12	$\frac{k}{20} = \frac{5}{k} \Rightarrow k = \pm 10$ (1); $k = 10$ gives coincident lines, so $k = -10$ (1).	1+1
13	Adding: $14(x+y) = 124$ so $x + y = \frac{62}{7}$ (1); subtracting: $x - y = 2$, so $x = \frac{38}{7}$, $y = \frac{24}{7}$ (1).	1+1
14	x-intercept $(3, 0)$, y-intercept $(0, 5)$ (1); area $= \frac{1}{2} \times 3 \times 5 = \frac{15}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $3p + 2q = \frac{19}{10}$, $4p + 5q = 3$ (1); eliminating q: $(7)p = \frac{7}{2}$ so $p = \frac{1}{2}$ (1); $q = \frac{1}{5}$ (1); $x = 2$, $y = 5$ (1).	1+1+1+1
16	Need $\frac{a+b}{5} = \frac{a-b}{1} = \frac{18}{9} = 2$ (1); $a + b = 10$, $a - b = 2$ (1); $a = 6$, $b = 4$ (1).	1+1+1
17	Intersection $(3, 2)$ (1); x-intercepts 9 and 4 (1); area $= \frac{1}{2} \times 5 \times 2 = 5$ (1).	1+1+1
	OR: Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $5p + 2q = \frac{3}{2}$, $3p + 4q = \frac{11}{6}$ (1); eliminating q: $(14)p = \frac{7}{3}$ so $p = \frac{1}{6}$ (1); $q = \frac{1}{3}$ (1); $x = 6$, $y = 3$ (1).	1+1+1+1
18	Need $\frac{a+b}{3} = \frac{a-b}{1} = \frac{8}{2} = 4$ (1); $a + b = 12$, $a - b = 4$ (1); $a = 8$, $b = 4$ (1).	1+1+1
	OR: Intersection $(3, 2)$ (1); x-intercepts 6 and 5 (1); area $= \frac{1}{2} \times 1 \times 2 = 1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + y = 9$ (1). (ii) $(10x + y) - (10y + x) = 27 \Rightarrow 9(x - y) = 27 \Rightarrow x - y = 3$ (1). (iii) Adding: $2x = 12$, $x = 6$, $y = 3$; the number is 63 (1). (iv) $9(x - y) = 36 \Rightarrow x - y = 4$; with $x + y = 9$ we get $x = 6.5$, not a digit, so no such number exists (2).	1+1+1+2
20	Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $270p + 270q = \frac{21}{2}$, $360p + 180q = 11$ (2); $p = \frac{1}{45}$, $q = \frac{1}{60}$ (2); train 45 km/h, bus 60 km/h (1).	2+2+1
	OR: Intersection $(\frac{31}{7}, -\frac{2}{7})$ (2); y-intercepts $(0, 13)$ and $(0, -\frac{5}{2})$ (1); base $= \frac{31}{2}$, height $= \frac{31}{7}$ (1); area $= \frac{1}{2} \times$ base $\times$ height $= \frac{961}{28}$ sq. units (1).	2+1+1+1

Mistakes to avoid:

- Applying the ratio test with the wrong pairing, e.g. comparing $\frac{a_1}{b_2}$ instead of $\frac{a_1}{a_2}$.
- Saying a pair with $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ always has no solution; check $\frac{c_1}{c_2}$ to separate coincident lines from parallel lines.
- Subtracting only some terms in elimination, or forgetting to multiply the right-hand side as well as the left.

- Reading the result $0 = 0$ as 'no solution'; it signals infinitely many solutions, while $0 = 5$ signals none.
- Not substituting back to find the second variable, or stating only x when the question asks for both.
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